

ISRO TECHNICAL ASSISTANT

COMPUTER SCIENCE & ENGINEERING

(Written Test Syllabus)

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1. Mathematics:

- **Matrices,**
- **Trigonometry,**
- **Complex Numbers,**
- **Analytical Geometry,**
- **Differentiation and its Applications,**
- **Integration and its Applications,**
- **Differential Equations**

2. Computer organizations & Architecture:

- **Basic Structure Of Computers,**
- **Machine Instructions and Programs,**
- **Basic Processing Unit,**
- **Input/Output Organization,**
- **The Memory system,**
- **Processors and Pipelining,**
- **Computer Function and Internal Memory,**
- **External Memory and Input/Output,**
- **Processor Structure, Control Unit Organization**

3. Programming:

- Programming in C,
- Object oriented programming concepts including classes, Polymorphism, Inheritance, and
- Programming in C++, and Java

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4. Data Structures:

- **Pointers and Dynamic Memory allocation, Files,**
- **Introduction to data structures & Linked lists,**
- **Stack & Queues,**
- **Trees,**
- **Sorting, Searching and**
- **Application of Data Structures**

5. Design and Analysis of Algorithms:

- Algorithm complexity,
- Brute Force, Algorithms Design Techniques — Divide and Conquer, Greedy Method,
- Dynamic Programming, Backtracking, Branch and Bound, NP-Hard and NP Complete Problems

6. Principles of Programming Languages:

- BNF, Variables,
- Data Types,
- Control Structures,
- Scope and Extent,
- Data Abstraction,
- Concurrency concepts,
- Exception Handling,
- Functional Programming, and
- Logic Programming

7. Database Management Systems:

- Database system, Database Design,
- Structured Query Language,
- Normalizations and Transactions

8. Operating Systems:

- Introduction to operating systems,
- Process management,
- Synchronization & Deadlocks,
- Memory management,
- Virtual memory management,
- File system

9. Linux System Administration :

- Introduction to Linux Administration,
- Understanding Linux system organization,
- Working with files & directories,
- File permissions & file utilities,
- vi editor,
- Expressions & search patterns,
- Linux Environment,
- Shell programming

10.Digital Electronics:

- Digital Principles,
- Boolean algebra & Logic gates,
- Combinational Circuits,
- Sequential Circuits,
- A/D, D/A,
- Memory and Programmable Logic,
- Introduction to Computers & Computer software,
- Peripherals & Memory

11. Computer Network:

- Introduction to Data Communication,
- Networking and Topologies,
- OSI Model & TCP/IP Model
- Flow Control & Access Control
- Error Detection and Error Correction,
- LAN Components and Protocols,
- Network Addressing, Classless & Classfull Networks
- Concept of Subnetting
- Wide Area Networks

12. Software Engineering:

- The software problem & Software process,
- Software requirements analysis and specifications,
- Software architecture & planning a software project,
- Various SDLC Models and their phases
- Design, Coding and Testing,
- Testing methodology, Testing techniques, Test automation and testing tools, Debugging process
- Software Project Management
- Devops

13. Network Security:

- Introduction to Computer Security & Cryptography,
- User Authentication & Access Control,
- Intrusion Detection and Malicious Software,
- Denial of Service and Firewall,
- Wireless Security, web Security, Security and Law,
- Internet Governance and Email Policy

14. Information Storage and Management:

- Introduction to Information Storage and Data centre Environment,
- Data Protection : RAID, Intelligent Storage System, Storage Area Networks, Network Attached Storage

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15. Web Programming :

- Fundamentals and Introduction to XHTML,
- Java Script and XHTML Documents and Dynamic documents with JavaScript,
- Introduction to XML,
- Introduction to PHP,
- Database Access through Web,
- Java Web Software

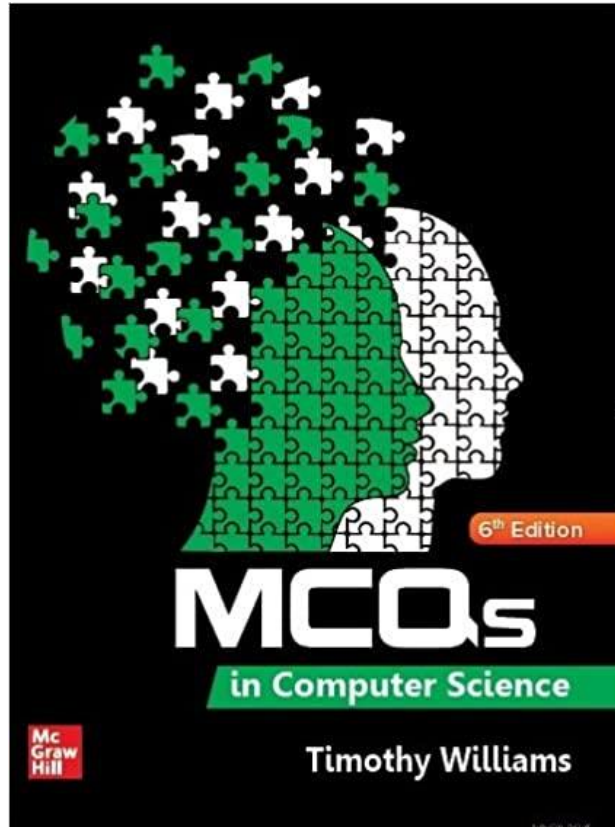
16. Microprocessor & Micro Controller:

- Fundamentals of Microprocessor 8085 & 8086,
- Architecture of 8085 & 8086,
- Architecture of Microcontrollers 8051,
- Programming concepts in 8085 & 8051,
- Application of 8085 & 8051,
- Microcontrollers and its interfacing,
- AVR microcontrollers fundamentals

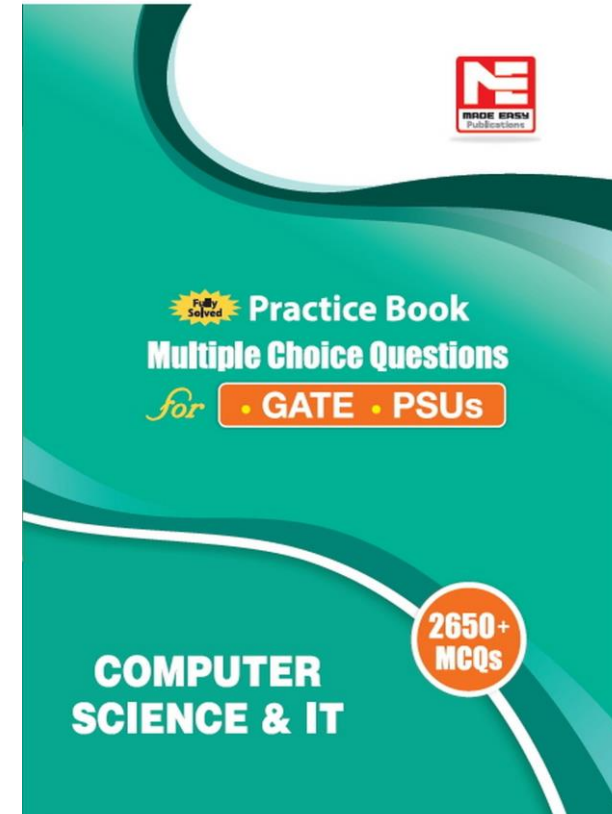
17. Extras:

- 1 or 2 questions are from Applied Physics
- 2 or 3 questions from Basic electrical & electronics engineering
- 2 to 3 questions from Cloud computing & Big data analytics

Recommended Books



Amazon - <https://amzn.to/3NTpzlR>



Amazon - <https://amzn.to/44JCEnP>

Some Advise for written test:

- Try to solve minimum 65 to 70 (out of 80) question in written test (50 question out of 60).
- If you are getting around 70 % marks then more chance to clear the written test.
- As you know final selection will be only on basis of written test marks -> then only toppers will be selected.
- Try to solve all previous questions and analyze then stick with syllabus, as knowing the what not to study is more important.
- Try to solve as much as numerical and objective question in home.
- First you top in written test , then think of Skill test , written test is more important than skill test for selection.

Telegram Group

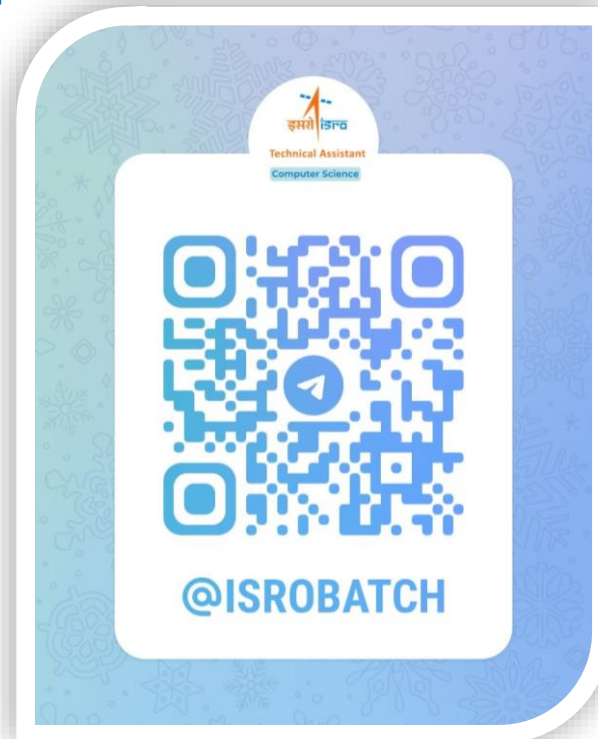
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